THE PICARD FUNCTOR IS AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file quot.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Theorem bootstrap-theorem-final-bootstrap (canonical tag 04S6). cited at proof line 3251. Bootstrap descent of algebraicity.
- Proposition proposition-pic (canonical tag 0D04). cited at proof line 3257. Picardstack algebraic.
- Lemma lemma-compare-pic-with-section (canonical tag 0D2A). cited at proof line 3259. Sectioned substack over Picardstack.
- Lemma algebraic-lemma-representable-morphism-to-algebraic (canonical tag 05UM). cited at proof line 3261. Transfer of algebraicity.
- Lemma lemma-flat-geometrically-connected-fibres-with-section-functor-stack (canonical tag 0D2B). cited at proof line 3264. Kernel functor algebraic.
- Lemma algebraic-lemma-characterize-representable-by-space (canonical tag 02ZX). cited at proof line 3266. Representability criterion.
- Lemma lemma-flat-geometrically-connected-fibres-with-section (canonical tag 0D28). cited at proof line 3269. Kernel = Picard functor.

Source reference. Original theorem 04S6: sources/stacks-chow/tags.txt, source lines 6209–6209. Cited at proof line 3251: descent of algebraicity along the representable cover B' -> B.

Source reference. Original proposition 0D04: sources/stacks-chow/tags.txt, source lines 16299–16299. Cited at proof line 3257: Picardstack algebraic.

Source reference. Original lemma 0D2A: sources/stacks-chow/tags.txt, source lines 16375–16375. Cited at proof line 3259: the morphism to the Picard stack.

Source reference. Original lemma 05UM: sources/stacks-chow/tags.txt, source lines 7521–7521. Cited at proof line 3261: sectioned substack algebraic.

Source reference. Original lemma 0D2B: sources/stacks-chow/tags.txt, source lines 16376–16376. Cited at proof line 3264: kernel functor algebraic space.

Source reference. Original lemma 02ZX: sources/stacks-chow/tags.txt, source lines 3928–3928. Cited at proof line 3266: kernel functor is an algebraic space.

Source reference. Original lemma 0D28: sources/stacks-chow/tags.txt, source lines 16373–16373. Cited at proof line 3269: functor identity.

Date: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proposition 0.2. *Let S be a scheme. Let $f : X \to B$ be a morphism of algebraic spaces over S. Assume that*

(1) *f is flat, of finite presentation, and proper, and*
(2) *$\mathcal{O}_T \to f_{T,*}\mathcal{O}_{X_T}$ is an isomorphism for all schemes T over B.*

Then $\mathrm{Pic}_{X/B}$ is an algebraic space.

Proof. There exists a surjective, flat, finitely presented morphism $B' \to B$ of algebraic spaces such that the base change $X' = X \times_B B'$ over B' has a section: namely, we can take $B' = X$. Observe that $\mathrm{Pic}_{X'/B'} = B' \times_B \mathrm{Pic}_{X/B}$. Hence $\mathrm{Pic}_{X'/B'} \to \mathrm{Pic}_{X/B}$ is representable by algebraic spaces, surjective, flat, and finitely presented. Hence, if we can show that $\mathrm{Pic}_{X'/B'}$ is an algebraic space, then it follows that $\mathrm{Pic}_{X/B}$ is an algebraic space by Bootstrap, Theorem 04S6. In this way we reduce to the case described in the next paragraph.

In addition to the assumptions of the proposition, assume that we have a section $\sigma : B \to X$. By Proposition 0D04 we see that $\mathcal{P}ic_{X/B}$ is an algebraic stack. By Lemma 0D2A and Algebraic Stacks, Lemma 05UM we see that $\mathcal{P}ic_{X/B,\sigma}$ is an algebraic stack. By Lemma 0D2B and Algebraic Stacks, Lemma 02ZX we see that $T \mapsto \mathrm{Ker}(\sigma_T^* : \mathrm{Pic}(X_T) \to \mathrm{Pic}(T))$ is an algebraic space. By Lemma 0D28 this functor is the same as $\mathrm{Pic}_{X/B}$. □